

Code and submit the full solution with source codes.

This assessment contains 1 Tasks only with 3 basic and 2 advance questions.

Candidate must complete the basic sections.

Candidate able to complete the advance section as well will be ranked higher.

Code and submit the full solution with source codes.

Grading criteria:

1. *Correctness*
 - *Comprehension - understand the requirement and produce code accordingly.*
 - *Bug free - the coding is done without logic bugs.*
 - *Performant - able to execute within acceptable timeframe.*
2. *Style*
 - *Code formatting - spaces, indentation, braces, new lines.*
 - *Naming convention - variables, methods, classes are named meaningfully.*
 - *Readability - clear and organized logic flow, easy to read and maintain by others.*
3. *Comprehensiveness*
 - *Methods/Functions/Classes/Designs are flexible; cater for reusability and future extension.*
 - *Has proper validation and able to handle case scenarios.*
 - *Has unit test(s), if possible.*

Task:

Question 1 (Basic):

Use **.NET 6.0 or later** to develop a REST API function which receives transaction information from **allowed Partner** to further process the transaction information. This request will return a success / failed result with result messages back to **allowed Partner**.

Partner No	Allowed Partner	Partner's Password
FG-00001	FAKEGOOGLE	FAKEPASSWORD1234
FG-00002	FAKEPEOPLE	FAKEPASSWORD4578

API Specifications

Receiving Component	[Domain]/api/submitttrxmessage
Purpose	To receive transaction from partner for further processing
Type of Interface	HTTP POST to RESTful API (JSON)
Frequency (or periodicity)	On demand
Triggering Event	Triggered by partner site or application

Request Message

body

This object is used to describe the transaction details of the vertical partner

No	Field Name	Type	Size	Req	Description
1	partnerkey	String	50	M	The allowed partner's key
2	partnerrefno	String	50	M	Partner's reference number for this transaction Sample: - FG-00001
3	partnerpassword	String	50	M	Encode to Base64 format.
4	totalamount	Long	-	M	Total amount of payment include discount in the MYR only. Business Logic: - Only allow positive value - value in cents Example: 1000 = MYR 10.00
5	items	Array of itemdetail	-	O	Array of items purchased through this transaction
6	timestamp	String		M	String representation of the UTC time of the operation in ISO 8601 format ie. 2024-08-15T02:11:22.000000Z
7	sig			M	Message Signature. Sort the parameters alphabetically by their name, except for timestamp and sig. sigtimestamp format are yyyyMMddHHmmss Concatenate all parameter values into a single string in the specified order, encode it using SHA-256 with UTF-8 to produce a lowercase hexadecimal hash , and then convert the resulting hash to Base64 using UTF-8 encoding .

itemdetail definition (items)

This object is used to describe the items information

No	Field Name	Type	Size	Req	Description
1	partneritemref	String	50	M	Reference ID the partner uses for this item Business Logic: - Cannot be null or empty
2	name	String	100	M	Name of the item Business Logic: - Cannot be null or empty
3	qty	Integer	-	M	Quantity of the item bought
4	unitprice	Long	-	M	Price of one unit of the item in the currency of the transaction Business Logic: - Only allow positive value - Quantity must not exceed 5 - value in cents - Example: 1000 = MYR 10.00

Response Message

body

No	Field Name	Type	Size	Req	Description
1	result	Integer	-	M	Whether the operation was successful or not. • 1 is successful, • 0 if errors encountered.
2	totalamount	Long	-	O	Total amount of payment include discount in the MYR only. Business Logic: - Only allow positive value - value in cents Example: 1000 = MYR 10.00
3	totaldiscount	Long	-	O	Total discount of payment in the MYR only. Business Logic: - Only allow positive value - value in cents Example: 1000 = MYR 10.00
4	finalamount	Long	-	O	Final amount that customer paid in the MYR only. Business Logic: - Only allow positive value - value in cents Example: 1000 = MYR 10.00
5	resultmessage	String	-	O	Result message if the operation was Failure

Sample Request

```
{
  "partnerkey": "FAKEGOOGLE",
  "partnerrefno": "FG-00001",
  "partnerpassword": "RkFLRVBBU1NXT1JEMTIzNA==",
  "totalamount": 1000,
  "items": [
    {
      "partneritemref": "i-00001",
      "name": "Pen",
      "qty": 4,
      "unitprice": 200
    },
    {
      "partneritemref": "i-00002",
      "name": "Ruler",
      "qty": 2,
      "unitprice": 100
    }
  ],
  "timestamp": "2024-08-15T02:11:22.0000000Z",
  "sig": "
MDE3ZTBkODg4ZDNhYzU0ZDBlZWVmNmU2NmUyOWRhZWU4Y2M1NzQ1OTIzZGRjYTc1ZGNjOTkwY
zg2MWJlMDExMw=="
}
```

Sample Response (Success)

```
{"result":1,"totalamount":1000,"totaldiscount":0,"finalamount":1000}
```

Sample Response (Failed)

```
{"result":0,"resultmessage":"Access Denied!"}
```

To generate the signature output based on instruction:

1. **Message Signature Parameter Order**
timestamp + partnerkey + partnerrefno + totalamount + partnerpassword(encoded)
Example: 20240815021122FAKEGOOGLEFG-000011000RkFLRVBBU1NXT1JEMTIZNA==
 2. **Apply SHA-256 hashing (UTF-8 encoded input)(lowercase hexadecimal hash output)**
017e0d888d3ac54d0eedf6e66e29daee8cc5745923ddca75dcc990c861be0113
 3. **Convert the SHA-256 hash to Base64 (UTF-8 encoding)**
MDE3ZTBkODg4ZDNhYzU0ZDBlZWVmNmU2NmUyOWRhZWU4Y2M1NzQ1OTIzZGRjYTc1ZGNjOTkwY
zg2MWJlMDExMw==
-

This API act as a middleware to validate the payload data only.

- a. Ensure all fields name matches the API Specifications above
- b. All fields must be validated before executing business logic
 - i. Example:
 1. Mandatory column cannot be NULL or not provided
 2. Positive value cannot accept **negative** value

NOTE: Read the conditions carefully for each field

Question 2 (Basic):

Apply the validation against the fields in the request and return them in error code. Below is an example error result messages we are looking for.

No	Error : Result Message	Description
1.	Access Denied!	Unauthorized partner or Signature Mismatch
2.	Invalid Total Amount.	Only applicable when itemDetails is provided. The total value stated in itemDetails array not equal to value in totalamount .
3.	Expired.	Provided timestamp exceed server time +-5min Example: server time is 13 Dec 2021 15:04:02 The valid time will be +-5 Min of the server time. Other than this will consider the API is expired.
4.	##ParamName## is Required.	Mandatory Parameter is not provided Example: partnerrefno is required.
5.	##Any Other Possible Messages##	If you able detect more validation, feel free to add in and provide your own error messages.

Question 3 (Basic):

In continuation of the **Question 1** API, you are tasked that calculates the final amount to be paid after applying several discount rules based on the **totalamount**. The business rules for applying discounts are as follows:

Discount Rules:

Base Discount:

If **totalamount** is less than MYR 200: No discount is applied.

If **totalamount** is between MYR 200 and MYR 500 (inclusive): Apply a 5% discount.

If **totalamount** is between MYR 501 and MYR 800 (inclusive): Apply a 7% discount.

If **totalAmount** is between MYR 801 and MYR 1200 (inclusive): Apply a 10% discount.

If **totalAmount** is greater than MYR 1200: Apply a 15% discount.

Conditional Discounts:

If **totalAmount** is a prime number above MYR 500: Apply an additional 8% discount.

If **totalAmount** ends in the digit 5 and is above MYR 900: Apply an additional 10% discount.

Cap on Maximum Discount:

The total discount (base + conditional) cannot exceed 20% of the **totalamount**.

Sample calculation 1:

totalamount: 100000

Base Discount:

Meets the condition of being within MYR 801 and MYR 1200: **10% discount**

Conditional Discounts:

None of the condition are met.

Cap on Maximum Discount:

The total discount is 10%, which is below the 20% threshold, so a 10% discount will be applied.

Total Discount: RM1000 x 10% = RM100

Your Sample 1 Response will be

```
{ "result":1, "totalamount":100000, "totaldiscount":10000, "finalamount":90000
}
```

Sample calculation 2:

totalamount: 120500

Base Discount:

Meets the condition of being greater than MYR 1200: **15% discount**

Conditional Discounts:

Meets the condition of ending in the digit 5 and being above MYR 900: **additional 10% discount**

Cap on Maximum Discount:

The total discount is 25%, which is above the 20% threshold, so a 20% discount will be applied.

Total Discount: RM1205 x 20% = RM241

Your Sample 1 Response will be

```
{ "result":1, "totalamount":120500, "totaldiscount":24100, "finalamount":96400
}
```

Question 4 (Advance):

In continuation of the **Basic** scenarios, saved all the request body, response body and possible logs in text file to ease future troubleshooting **using log4net**. Ensure that passwords in the log file are encrypted for security.

Question 5 (Advance):

Containerizing the application using **Docker** and provide Docker File.